

Module 1: Process Models & Agile Methodologies

Waterfall, Spiral, Scrum Sprints, Extreme Programming & RMMM

Module I: Process Models & Agile Methodologies — Topics Covered

1. Software Crisis & Layered Technology

2. Prescriptive Lifecycle Models (Waterfall, Spiral, RAD)

3. Agile Manifesto & Scrum Framework

4. Extreme Programming (XP & TDD)

5. Project Scheduling & Risk Management (RMMM)

1. Software Process Lifecycle Models Comparison

Model	Core Characteristics	Ideal Project Profile
Waterfall Model	Strict linear sequential phases. High documentation discipline. No customer feedback until late testing phase.	Well-understood, stable requirements (e.g., compilers, aerospace flight control).
Spiral Model (Boehm)	Risk-driven evolutionary meta-model combining prototyping with systematic waterfall milestones across 4 quadrants.	Large, expensive, high-risk, mission-critical systems.
Scrum (Agile)	Iterative, time-boxed Sprints (2-4 weeks), daily stand-ups, self-organizing teams, shippable product increments.	Rapidly evolving requirements, web/mobile applications.